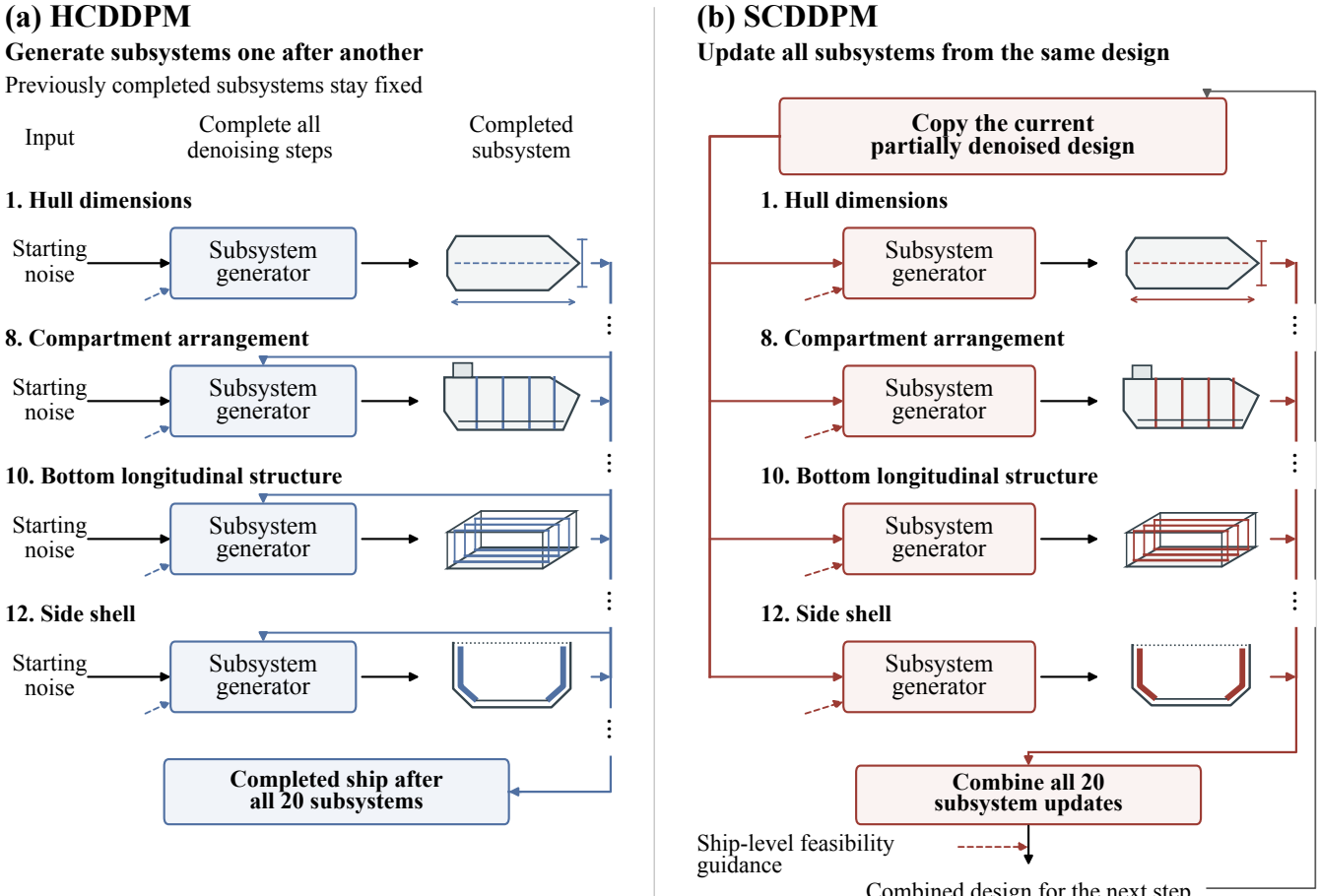
(a) HCDDPM
Generate subsystems one after another
Previously completed subsystems stay fixed
Input
Complete all denoising steps
Completed subsystem
1. Hull dimensions
Starting noise
Subsystem generator
8. Compartment arrangement
Starting noise
Subsystem generator
10. Bottom longitudinal structure
Starting noise
Subsystem generator
12. Side shell
Starting noise
Subsystem generator
Completed ship after all 20 subsystems
(b) SCDDPM
Update all subsystems from the same design
Copy the current partially denoised design
1. Hull dimensions
Subsystem generator
8. Compartment arrangement
Subsystem generator
10. Bottom longitudinal structure
Subsystem generator
12. Side shell
Subsystem generator
Combine all 20 subsystem updates
Ship-level feasibility guidance
Combined design for the next step
Dashed arrows to generators: subsystem-level feasibility guidance